Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2023	ENGI3311-WE01

Title: (BEng/MEng)

Geotechnics 3

Time Allowed:	2 hours
Additional Material provided:	Yes - Formula Sheet included as part of exam paper.
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Q1

(a) A typical lab procedure to obtain soil samples with different dry density is to apply a static load to mixtures of dry soil and water at constant gravimetric water content. Consider a soil sample that has a value of gravimetric water content of 0.8 and a specific gravity of 2.65. Determine the maximum possible value for the dry density of the sample that could be obtained from following this procedure. [25%]

(b) Figure Q1b illustrates a lake and an element of sand A at a depth of 10m below the horizontal ground bed of the lake. Initially, the water level in the lake is at 10m above the ground bed as indicated in Figure Q1b(left). Due to a heavy rainfall, the water level in the lake raises to 15m (see Figure Q1b(right)). Has the soil element A experienced any deformation after the rainfall? Justify your answer from first principles. The bulk unit weight of the sand is 20 kN/m^3. [25%]

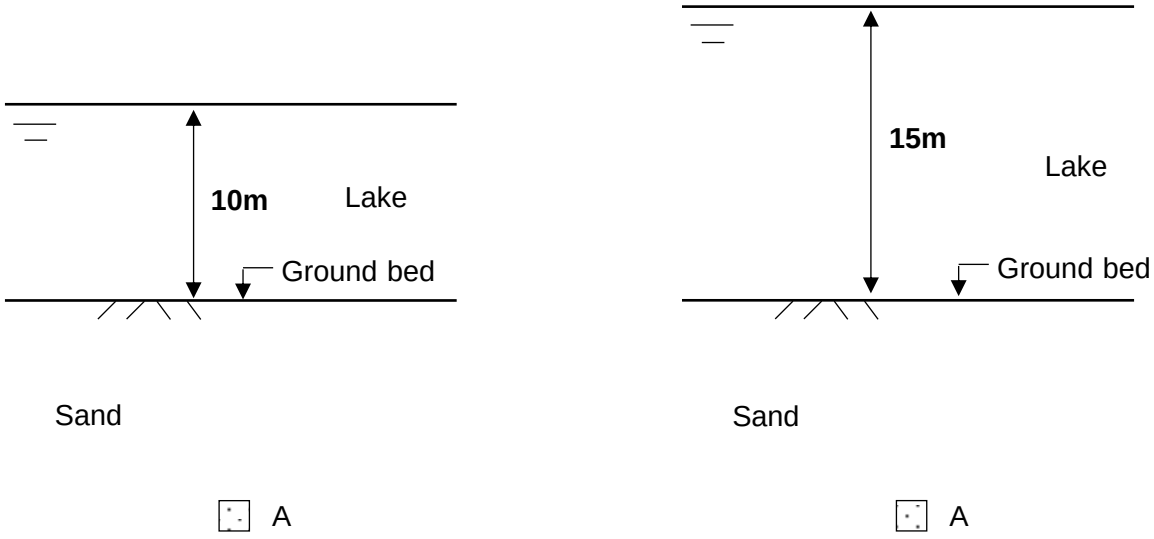


Figure Q1b (Not to scale)

(c) A different situation is illustrated in Figure Q1c where an element of sand B at a depth of 7m below the ground surface is now considered. Initially, the water table level is 2m below the ground surface (Figure Q1c(left)). Due to pumping, the water table lowers to 4m below the ground surface (Figure Q1c(right)). Does the element of soil at B experience any deformation due to the pumping? Justify your answer from first principles. The dry and saturated bulk unit weight of the sand are 16.5 and 20 kN/m^3, respectively. Consider that the height of capillary rise can be neglected. [25%]

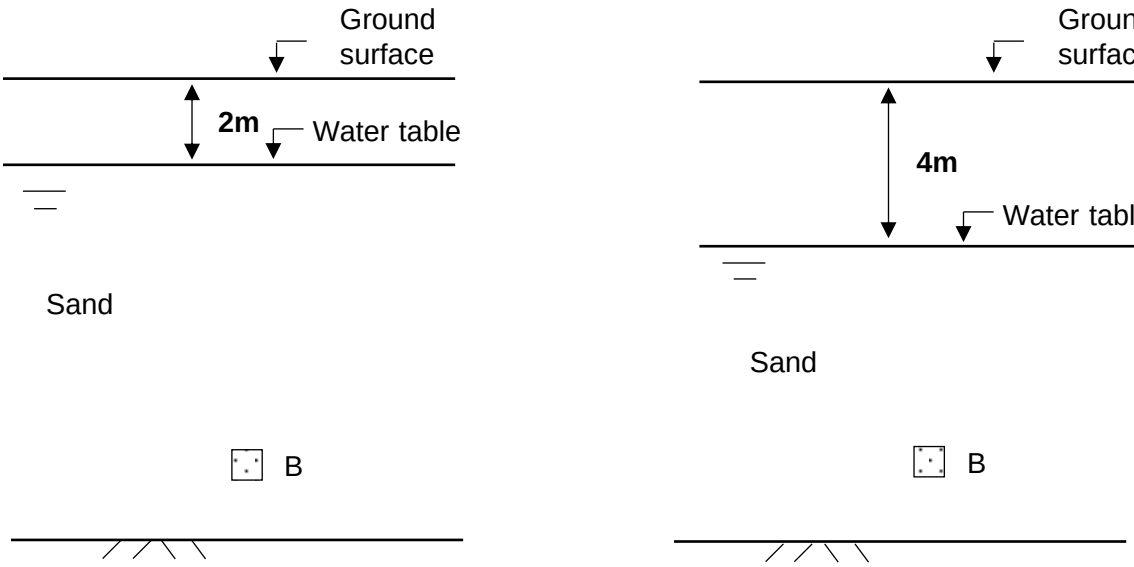


Figure Q1c (Not to scale)

(d) Consider a sedimentary clay that has been deposited under K_0 conditions with K_0=0.5. Assuming that the simplified stress conditions of the triaxial test are valid, plot the stress path in the *q*:*p*' plane (where *q* is the deviatoric stress and *p*' is the mean effective stress as defined in Cambridge's notation) that an element of soil P has experienced during its geological history from its formation to its current location 15m below the ground surface. Assume that the water table is always at the ground surface and that the bulk unit weight of the saturated clay is 20 kN/m^3. [25%]

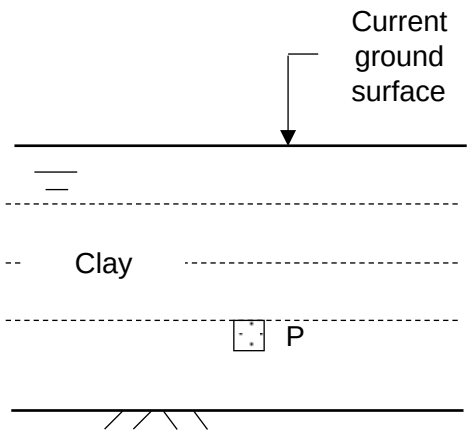


Figure Q1d (Not to scale)

continued

Q2

In the proximities of a coastline, the soil profile can be approximated as that shown in Figure Q2a.

A borehole at 500m from the coast indicates that the water table (or phreatic level) is 50m above an approximately horizontal bedrock which can be considered impermeable. This bedrock is located at 40m below the sea level (see Figure Q2a). From the borehole information, the permeability k is estimated as 10^{-5} m/s.

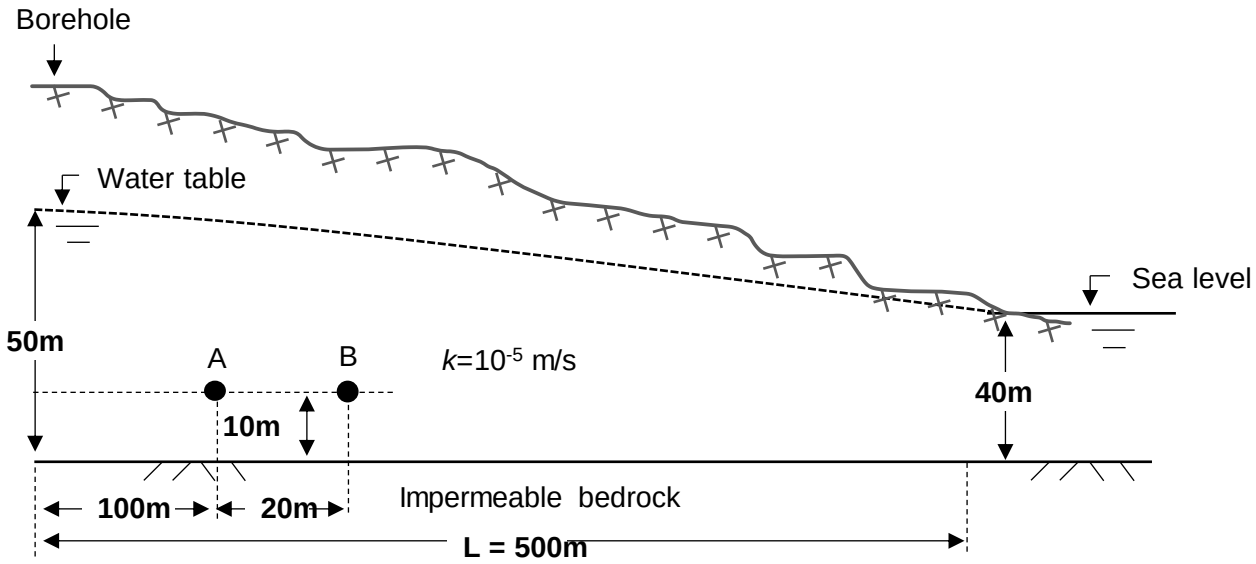


Figure Q2a (Not to scale)

Assuming that the Dupuit's hypotheses are valid,

a) Find the equation that represents the phreatic level. Justify all your assumptions and calculations. [15%]
b) Find the volumetric flow rate (per meter run) of water discharged to the sea. [15%]
c) Estimate the values of pore water pressure at A and B (see Figure Q2a). [10%]

A large underground train station is to be constructed as indicated in Figure Q2b. The geotechnical consulting group involved in the construction wants to study the effect that the station will have on the flow of water to the sea, as well as on the phreatic level. The concrete walls, floor and roof of the station can be considered impermeable (see Figure Q2b).

continued

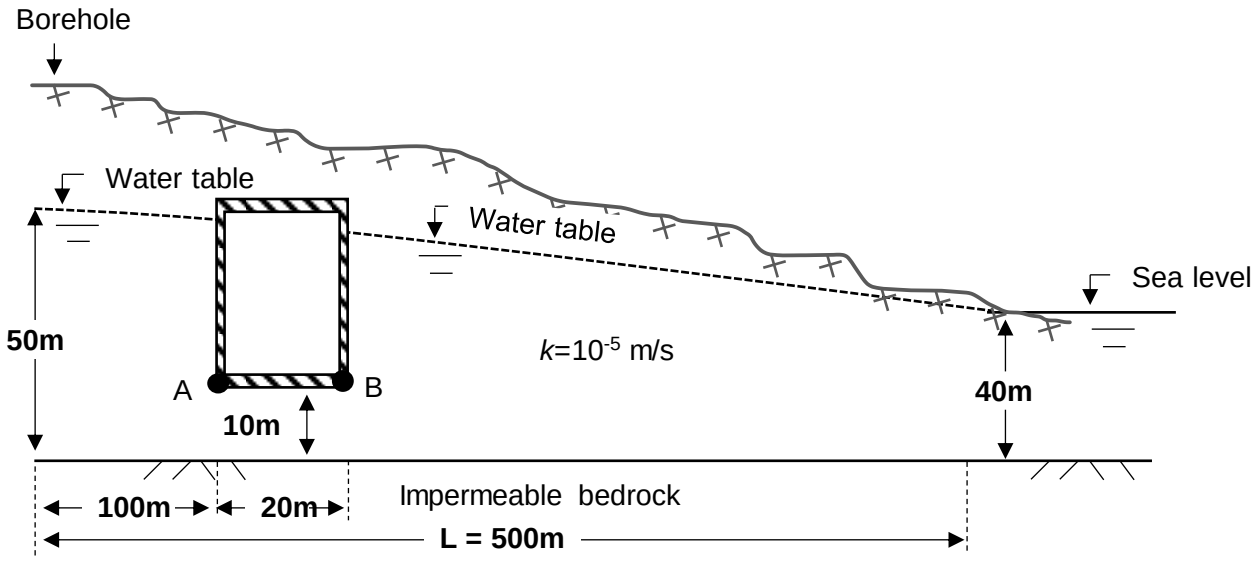


Figure Q2b (Not to scale)

Assuming that the phreatic level at 500m from the sea (and at 50m above the bedrock) is unaffected by the construction of the station, find:

d) A suitable approach to calculate the phreatic level and volumetric flow rate discharged to the sea. Justify your answer providing sketches. [15%]

e) Estimate the variation in hight of the phreatic level at points A and B. [25%]

f) Estimate the values of pore water pressure at points A and B (located at the floor of the station, see Figure Q2b) and compare them against the values obtained at c). [10%]

g) Estimate the new volumetric flow rate (per meter run) of water discharged to the sea and compare it against the value obtained in b). [10%]

Note: Assume also that Dupuit's hypotheses are still applicable.

Q3

A retaining gravity wall is constructed on a site to provide stability to a sand and clay fills (see Figure Q3). The site geotechnical engineer wants to carry out a preliminary stability assessment of the retaining wall against overturning based only on its factor of safety. The total height of the wall is 12.7m, its base is 10m and its top width is 4m, as indicated in Figure Q3. The fill material consists of a top 8m layer of sand with a saturated unit weight of 20kN/m^3 and a friction angle of 30°. This sand layer rests on a 4m layer of clay with a saturated unit weight of 20kN/m^3, a friction angle of 20°and a cohesion of 20kPa. The unit weight of the concrete in the wall is 23kN/m^3. The ground water level behind the wall is at the ground surface (see Figure Q3). The wall carries a variable uniform surcharge *q* of 30 kPa.

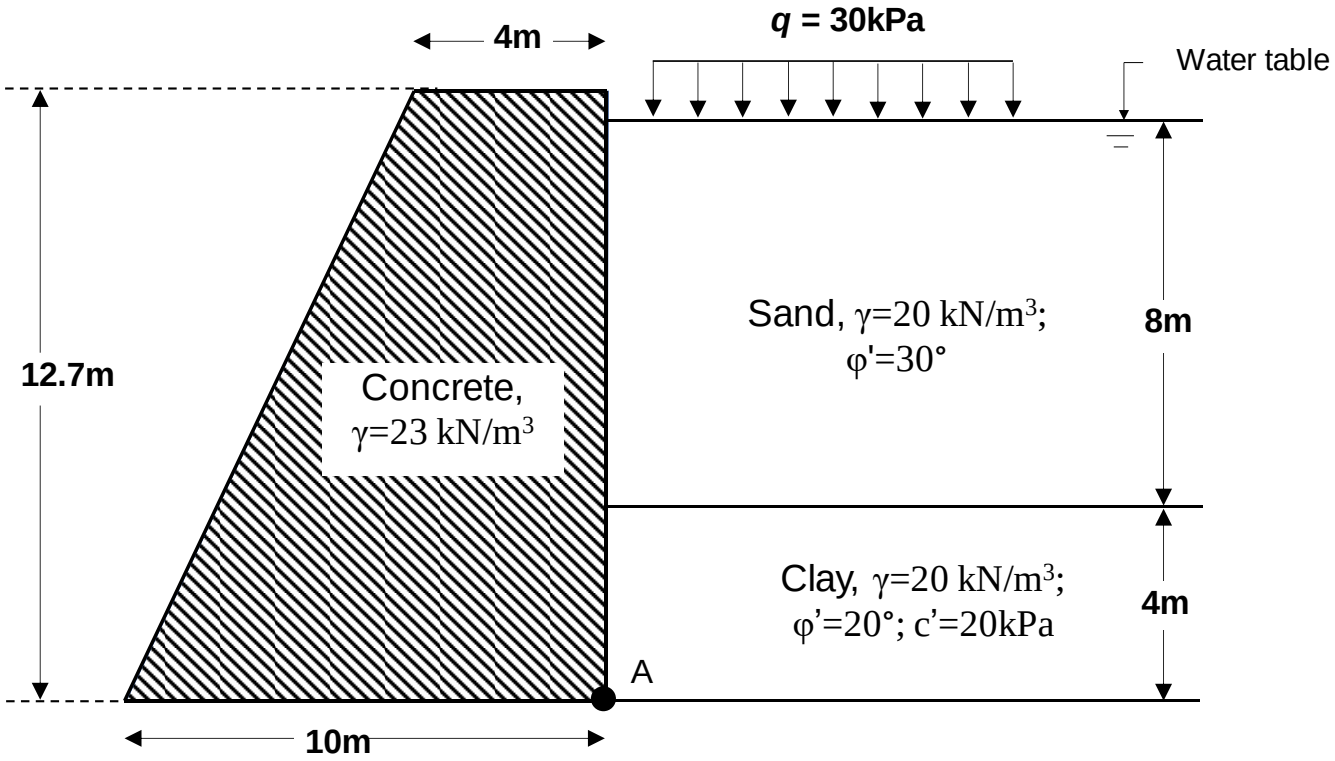


Figure Q3

(a) Assuming no friction between the wall and both soils, calculate the active earth pressure coefficients for each soil. [10%]
(b) Calculate the distribution of vertical total stress behind the wall. [10%]
(c) Calculate the distribution of vertical effective stress behind the wall. [10%]
(d) Calculate the distribution of horizontal effective stress behind the wall. [10%]
(e) Calculate the horizontal total stress and corresponding thrusts acting on the wall. [20%]
(f) Sketch the distribution of horizontal total stress against the wall, the corresponding thrusts and their respective positions. [20%]
(g) Check the stability against overturning based on the calculation of the factor of safety. [20%]

continued

Phase relationships

$e=\frac{V_v}{V_s}$	$S_r=\frac{V_w}{V_v}$	$v=1+e$	$n=\frac{e}{1+e}$	$w=\frac{M_w}{M_s}$
$\rho_s=G_s\rho_w$	$wG_s=S_re$	$\rho=\frac{M_s+M_w}{V_s+V_v}$	$\rho=\frac{(G_s+S_re)\rho_w}{1+e}$	$\gamma=\rho g$
$\gamma=\frac{(G_s+S_re)\gamma_w}{1+e}$	$\gamma'=\gamma_{sat}-\gamma_w$	$\rho_d=\frac{G_s\rho_w}{1+e}$	$\rho_d=\frac{\rho}{1+w}$	$\gamma_w\approx 10$ kN/m^3

Stresses and pore pressures in soils:

$\sigma_v=\rho gz=\gamma z$	$u_w=\rho_w gz_w=\gamma_w z_w$	$\sigma_v'=\sigma_v-u$	$u_a-u_w=\frac{2T}{R}$	$h_c=\frac{2T}{\gamma_w R}$

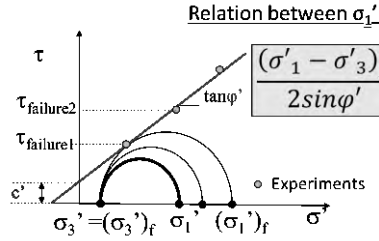
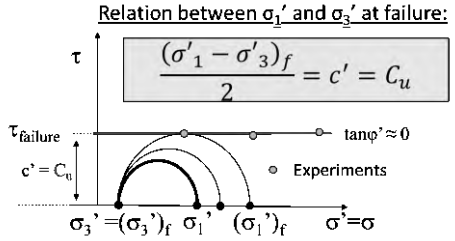
Elastic stress-strain relations:

The volumetric strain is: $\Delta\varepsilon_v=\Delta\varepsilon_1+\Delta\varepsilon_2+\Delta\varepsilon_3=\Delta\varepsilon_x+\Delta\varepsilon_y+\Delta\varepsilon_z$

$\Delta\varepsilon_x=\frac{1}{E'}\left(\Delta\sigma_x'-\nu'\Delta\sigma_y'-\nu'\Delta\sigma_z'\right)$	$\Delta\gamma_{xy}=\frac{\Delta\tau_{xy}}{G'}$	--
$\Delta\varepsilon_y=\frac{1}{E'}\left(\Delta\sigma_y'-\nu'\Delta\sigma_x'-\nu'\Delta\sigma_z'\right)$	$\Delta\gamma_{yz}=\frac{\Delta\tau_{yz}}{G'}$	$K'=\frac{E'}{3(1-2\nu')}$
$\Delta\varepsilon_z=\frac{1}{E'}\left(\Delta\sigma_z'-\nu'\Delta\sigma_y'-\nu'\Delta\sigma_x'\right)$	$\Delta\gamma_{zx}=\frac{\Delta\tau_{zx}}{G'}$	$G'=\frac{E'}{2(1+\nu')}$
$\Delta\varepsilon_v=\frac{\Delta p'}{K'}$		$\Delta\varepsilon_q^e=\frac{\Delta q}{3G'}$
$p'=\frac{\sigma'_1+\sigma'_2+\sigma'_3}{3}$		$q=\left[\frac{(\sigma_1'-\sigma_2')^2+(\sigma_2'-\sigma_3')^2+(\sigma_3'-\sigma_1')^2}{2}\right]^{1/2}$

where σ'_1, σ'_2 and σ'_3 are the principal effective stresses (i.e. $\sigma'_1\geq\sigma'_2\geq\sigma'_3$)

Failure envelopes:

$\tau=c'+\sigma'\tan\varphi'$ (Mohr-Coulomb)	$\tau=C_u$ (Tresca)
Relation between σ_1' and σ_3' at failure: $\frac{(\sigma'_1-\sigma'_3)_f}{2sin\varphi'}=\left[\frac{(\sigma'_1+\sigma'_3)_f}{2}+\frac{c'}{tan\varphi'}\right]$ 	Relation between σ_1' and σ_3' at failure: $\frac{(\sigma'_1-\sigma'_3)_f}{2}=c'=C_u$ 

Ground water flow:

$h=z+\frac{u}{\rho g}=z+\frac{u}{\gamma_w}$	$q_x=Aki_x=Ak\left(-\frac{\partial h}{\partial x}\right)$	$v_x=ki_x$	$v_{true,x}=v_x\frac{1}{n}$

Constant head permeameter:	Falling head permeameter:	Pumping test:
$q=Ak\frac{h_1-h_2}{L}$	$\ln\left(\frac{h_0}{h}\right)=\frac{Ak}{aL}t$	$\left(H^2-h^2\right)=\frac{q}{\pi k}\left(\ln R_0-\ln r\right)$

Laplace's Equation (3D)	Laplace's Equation (2D, flownets)	
$\frac{\partial^2 h}{\partial x^2}+\frac{\partial^2 h}{\partial y^2}+\frac{\partial^2 h}{\partial z^2}=0$	$\delta h=\frac{\Delta h}{N_d}$	$q=k\frac{N_f}{N_d}\Delta h;\ \ u=\gamma_w\left(h-z\right)$

continued

Critical hydraulic gradient:	$i_c = \dfrac{\gamma - \gamma_w}{\gamma_w}$

1D flow through an unconfined aquifer:	1D flow through a confined aquifer:
$q_x = \dfrac{k\left(H_1^2 - H_2^2\right)}{2L}$	$q_x = Aki_x = -Ak\dfrac{\partial h}{\partial x} = -k\dfrac{H_2 - H_1}{L}D$
$h = \sqrt{\dfrac{\left(H_2^2 - H_1^2\right)}{L}x + H_1^2}$	$h = \dfrac{H_2 - H_1}{L}x + H_1$

1D straining (oedometric test conditions):

1D normal compression line:	1D swelling line:
$e = e_1 - C_c \log_{10} \sigma'_v$	$e = e_2 - C_s \log_{10} \sigma'_v$

$OCR_{1D} = \dfrac{max.\ past\ value\ of\ \sigma'_v}{current\ value\ of\ \sigma'_v} = \dfrac{\sigma'_{vc}}{\sigma'_v}$	$\sigma'_h = K_0\sigma'_v$	$K_{0nc} \approx 1 - \sin\varphi'$	$K_0 \approx \left(1 - \sin\varphi'\right)\sqrt{OCR}$

Settlement of a very wide foundation:		
$\dfrac{\Delta H}{H_0} = \dfrac{e_0 - e_1}{1 + e_0}$	$\Delta H = \dfrac{H_0}{1 + e_0}\left[C_s \log_{10}\dfrac{\sigma'_{vc}}{\sigma'_{v0}} + C_c \log_{10}\dfrac{\sigma'_{v1}}{\sigma'_{vc}}\right]$	$\Delta\varepsilon_v = m_v\Delta\sigma'_v$

1D consolidation:		
$\dfrac{\partial u_e}{\partial t} = c_v\left(\dfrac{\partial^2 u_e}{\partial z^2}\right)$	$c_v = \dfrac{k}{m_v\gamma_w}$	$T_v = \dfrac{c_v t}{d^2}$

Axisymmetric conditions (triaxial test conditions):

$\Delta\varepsilon_v = \Delta\varepsilon_a + 2\Delta\varepsilon_r$	$\Delta\varepsilon_q = \dfrac{2}{3}\left(\Delta\varepsilon_a - \Delta\varepsilon_r\right)$
$p' = \dfrac{1}{3}\left(\sigma_a' + 2\sigma_r'\right)$	$q = \sigma_a - \sigma_r$

where σ'_a and σ'_r are the principal effective axial and radial stresses respectively

Iso-normal compression line:	Iso-swelling line:
$v = N - \lambda \ln p'$	$v = v_\kappa - \kappa \ln p'$

Critical states:	
$q = Mp'$	$v = \Gamma - \lambda \ln p'$

where $\Gamma = N - (\lambda - \kappa)\ln(2)$;

Modified Cam Clay yield curve *f*:		$q^2 - M^2 p'(p'_0 - p') = 0$	
Plastic volumetric strains	$\Delta\varepsilon_v^p = \dfrac{(\lambda - \kappa)}{v} ln\left(\dfrac{p'_{0B}}{p'_{0A}}\right)$	Plastic deviatoric strains	$\Delta\varepsilon_q^p = \dfrac{2\eta}{M^2 - \eta^2}\Delta\varepsilon_v^p$
Stress ratio		$\eta = \dfrac{q}{p'}$	
Relation between friction angle and critical state gradient:		$M = \left.\dfrac{q}{p'}\right\|_{at\ critical\ state} = \dfrac{6\sin\varphi'}{3 - \sin\varphi'}$	

continued

Overconsolidation ratio (OCR in terms of p')	$OCR = \frac{max.\ past\ value\ of\ p'}{current\ value\ of\ p'} = \frac{p'_0}{p'}$

Partial factors Eurocode 7 & factor of safety: (Bond & Harris, 2008; *Decoding EC7*)

$q_{Ed} \le q_{Rd}$ where, q_{Ed} is the design pressure and q_{Rd} is the design resistance	CASE	ACTIONS			Soil properties	
		Permanent		Variable	tan φ', c'	C_u
		Unfavourable	Favourable	Unfavourable		
	C	1.00	1.00	1.30	1.25	1.4

Factor of safety, FoS:	$FoS = \frac{available\ shear\ force,\ T_f}{allowable\ working\ load,\ T}$; $FoS = \frac{available\ bearing\ capcity,\ \tau_f}{allowable\ bearing\ capacity,\ \tau}$

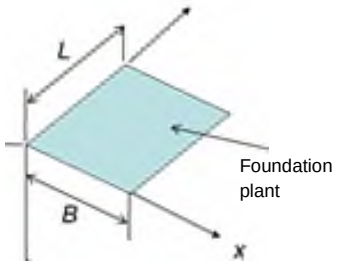
Shallow foundation settlement, δ:

Cohesionless soils:

Burland & Burbage:

$$\delta\ (mm) = f_s \cdot f_l \cdot f_t \cdot \left[\left(q'_n - \frac{2}{3}p'_0\right) \cdot B^{0.7} \cdot I_c\right]$$

where,



$f_s = \left(\frac{1.25 \cdot L/B}{\frac{L}{B} + 0.25}\right)^2$	$f_l = \frac{H}{z_l}\left(2 - \frac{H}{z_l}\right)$	$f_t = 1 + R_3 + R \cdot log\left(\frac{t}{3}\right)$

Cohesive soils: $\delta = \delta_i + \delta_c$ where,

$\delta_i\ (mm) = \mu_1 \cdot \mu_0 \cdot q_n \frac{B}{E_u}$	$\delta_c\ (mm) = \mu_g \cdot \delta_{oed}$ where, $\delta_{oed}\ (mm) = m_v \cdot \sigma'_v \cdot H$

Bearing capacity:

Brinch-Hansen (drained all soils)

$$q_f = c' \cdot N_c \cdot s_c + p'_{ov} \cdot N_q \cdot s_q + \frac{1}{2} \cdot \gamma' \cdot B \cdot N_y \cdot s_y$$

where,

$N_q = e^{\pi \tan \varphi'} \cdot \left(\tan(45 + \varphi'/2)\right)^2$	$s_q = 1 + (B'/L') \cdot \sin \varphi'$ (rectangular); $s_q = 1 + \sin \varphi'$ (circular and square)
$N_c = \left(N_q - 1\right) \cdot \cot \varphi'$	$s_c = \left(s_q \cdot N_q - 1\right)/\left(N_q - 1\right)$ (all)
$N_y = 2 \cdot \left(N_q - 1\right) \cdot \tan \varphi'$	$s_y = 1 - 0.3 \cdot (B'/L')$ (rectangular; $s_y = 0.7$ (circular and square)

depth factors are not permitted within EC7; load inclination, base and ground factors ignored.

Fine-grained soils (undrained clays; less accurate approximation if used for sands and gravels)

$q_f = 5.14 \cdot C_u \cdot s'_c$ where,

$s'_c = 1 + 0.2 \cdot \left(\frac{B'}{L'}\right)$ (rectangular shape; for circular and square shape B'=L')

load inclination, base and ground factors are ignored.

Bearing capacity in piles:

$$Q_u = Q_s + Q_b - W_p$$

Cohesionless soils:	Cohesive soils:
$Q_s = q_s \cdot A_s = A_s \cdot k_s \cdot \sigma_{v0} \cdot \tan \delta$	$Q_s = q_s \cdot A_s = \alpha \cdot C_u \cdot A_s$
$Q_b = q_b \cdot A_b = N_q \cdot \sigma'_{v0} \cdot A_b$	$Q_b = q_b \cdot A_b = N_c \cdot C_{ub} \cdot A_b$

Passive and active pressures:

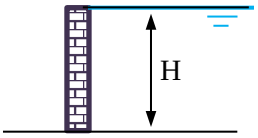
$\sigma'_h = K_a \cdot \sigma'_v$ (active)	$\sigma'_h = K_b \cdot \sigma'_v$ (passive)

continued

RANKINE'S theory: H high of the layer of soil behind the wall retaining wall, φ' is the friction angle and c' is the cohesion

In the following, $P_a = P_a' + U$ and $P_p = P_p' + U$,

where U is the resultant of the pore water pressure.



$K_a = \tan^2\left(\frac{\pi}{4} - \frac{\varphi'}{2}\right) = \left(\frac{\cos\varphi'}{1+\sin\varphi'}\right)^2 = \frac{1-\sin\varphi'}{1+\sin\varphi'}$		$K_p = \tan^2\left(\frac{\pi}{4} + \frac{\varphi'}{2}\right) = \left(\frac{\cos\varphi'}{1-\sin\varphi'}\right)^2 = \frac{1+\sin\varphi'}{1-\sin\varphi'}$
Cohesionless soils:	Cohesive: (Undrained)	Cohesive: (drained)
$\sigma'_h = K_a \sigma'_v$	$\sigma_h = \sigma_v - 2C_u$	$\sigma_h' = \sigma_v' K_a - 2c'\sqrt{K_a}$
$P_a' = \frac{1}{2} K_a \gamma' H^2$	$P_a = \frac{1}{2}(\gamma H - 2C_u)\left(H - \frac{2C_u}{\gamma}\right)$	$P_a' = \frac{K_a \gamma'}{2}\left(H - \frac{2c'}{\gamma'\sqrt{K_a}}\right)^2$
$\sigma'_h = K_p \sigma'_v$	$\sigma_h = 2C_u + \sigma_v$	$\sigma_h' = \sigma_v' K_p + 2c'\sqrt{K_p}$
$P_p' = \frac{1}{2} K_p \gamma' H^2$	$P_p = \frac{H}{2}(\gamma H + 4C_u)$	$P_p' = \frac{1}{2} K_p \gamma' H^2 + 2c'\sqrt{K_p} H$

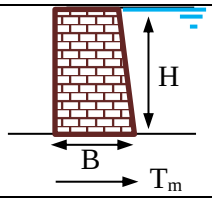
COULOMB'S theory: Friction and adhesion between soil and wall are φ_w and c_w, resp.

$K_a = \left(\frac{\cos\varphi'}{1+\sin\varphi'\sqrt{1+\tan\varphi_w'/\tan\varphi'}}\right)^2 \frac{1}{\cos\varphi_w'}$		$K_p = \left(\frac{\cos\varphi'}{1-\sin\varphi'\sqrt{1+\tan\varphi_w'/\tan\varphi'}}\right)^2 \frac{1}{\cos\varphi_w'}$
Cohesionless:	Cohesive: (Undrained)	Cohesive: (drained)
$\sigma'_h = K_a \sigma'_v$	$\sigma_h = \sigma_v - 2C_u\sqrt{1+\frac{C_w}{C_u}}$	$\sigma_h' = K_{a1}\sigma_v' - c' K_{a2}$
$P_a' = \frac{1}{2} K_a \gamma' H^2$	$P_a = \frac{\gamma}{2}\left(H - \frac{2C_u}{\gamma}\sqrt{1+\frac{C_w}{C_u}}\right)^2$	$P_a' = \frac{K_{a1}\gamma'}{2}\left(H - \frac{c' K_{a2}}{\gamma' K_{a1}}\right)^2$
$\sigma'_h = K_p \sigma'_v$	$\sigma_h = \gamma z + 2C_u\sqrt{1+\frac{C_w}{C_u}}$	$\sigma_h' = \gamma z K_{p1} + c' K_{p2}$
$P_p' = \frac{1}{2} K_p \gamma' H^2$	$P_p = \frac{H}{2}\left(\gamma H + 4C_u\sqrt{1+\frac{C_w}{C_u}}\right)$	$P_p' = \frac{1}{2} K_{p1}\gamma' H^2 + c' K_{p2} H$

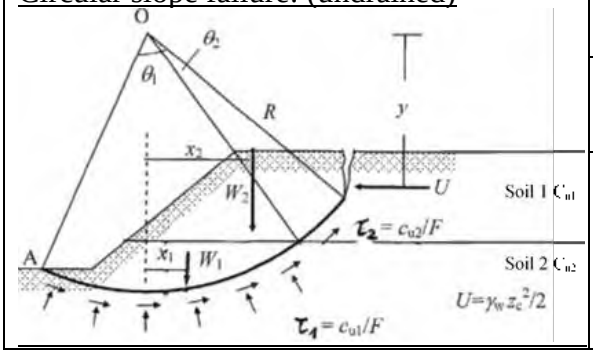
where, $K_{a1} = K_a$, $K_{p1} = K_p$ and

$K_{a2} = \frac{\cos\varphi'\left(2\sqrt{1+c_w'/c'} + (2+c_w'/c')\sin\varphi'\right)}{\left(1+\sin\varphi'\sqrt{1+\tan\varphi_w'/\tan\varphi'}\right)^2 \cos\varphi_w'}$	$K_{p2} = \frac{\cos\varphi'\left(2\sqrt{1+c_w'/c'} - (2+c_w'/c')\sin\varphi'\right)}{\left(1-\sin\varphi'\sqrt{1+\tan\varphi_w'/\tan\varphi'}\right)^2 \cos\varphi_w'}$
Cantilever wall in sand:	$FoS = \frac{N'\tan\varphi_w'}{T_m} = \frac{(W-U)\tan\varphi_w'}{T_m}$
Gravity wall in sand:	$FoS = \frac{N\tan\varphi_w'}{T_m}$

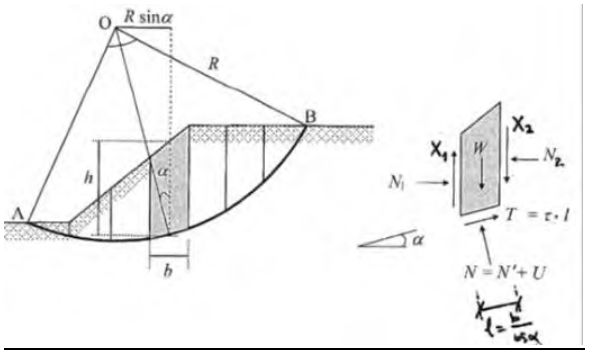
continued

Gravity wall in clay: 	(Undrained) $FoS = \frac{C_w B}{T_m}$	(Drained) $FoS = \frac{N'\tan\varphi_w' + c_w' B}{T_m}$

Slope stability: (Note: for undrained strength in infinite slope take c'=C_u and φ'=0)

Infinite slope with parallel flow: H: layer thickness, m: position of water table from bedrock, β: slope angle; γ_1 (dry soil; m=0), γ_2 (saturated soil; m=1).	$FoS = \frac{c' + H\left(\gamma_1(1-m) + m(\gamma_2 - \gamma_w)\right)\cos^2\beta\tan\varphi'}{H\left((1-m)\gamma_1 + m\gamma_2\right)\cos\beta\sin\beta}$
Vertical cut (dry soil):	$FoS = \frac{4C_u}{H\gamma\sin 2\alpha}$ (undrained)
	$FoS = \frac{4(c' + \sigma'\tan\varphi')}{H\gamma\sin 2\alpha}$ (drained)
Circular slope failure: (undrained) 	$FoS = \frac{R^2\theta C_u}{Wx}$
	$FoS = \frac{R^2\theta C_u}{Wx + Uy}$ (with cracks)
	$FoS = \frac{(\theta_1 C_{u1} + \theta_2 C_{u2})R^2}{W_1 x_1 + W_2 x_2 + Uy}$ (layered soil)

$\sin 2\alpha = 2\sin\alpha\cos\alpha$, $\cos 2\alpha = \cos^2\alpha - \sin^2\alpha$ and arc-length: $AB = R\theta$ (θ is in radians).

Circular: (drained): 	$FoS = \frac{\sum_{i=1}^{n} l\left(c' + \sigma'\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i} = \frac{\sum_{i=1}^{n}\left(lc' + N'\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i}$
Modified Bishop's b: width of the slice, l: length of the slice, α: inclination of the slice to the horizontal direction.	$FoS = \frac{\sum_{i=1}^{n}\left(bc' + (W - U\cos\alpha)\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i} M(\alpha)$ where, $M(\alpha) = \frac{1}{\cos\alpha\left[1 + \frac{\tan\alpha\tan\varphi'}{FoS}\right]}$
Fellenius:	$FoS = \frac{\sum_{i=1}^{n}\left(lc' + (W\cos\alpha - ul)\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i}$

END